Weakly Supervised Graph Model for DICOM-to-BIDS Inference

v3.0

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Abstract

We propose a hybrid heuristic-ML-graph pipeline for automated semantic labeling of heterogeneous clinical MRI DICOM series into BIDS format. The key innovation is joint prediction of three axes: (1) datatype (`anat`, `func`, `dwi`, `perf`, `fmap`), (2) suffix or coarse bucket (full BIDS suffixes for raw series; datatype-specific coarse buckets for derived maps), and (3) derived status (raw acquisition vs. derived map). Unlike previous approaches that excluded derived series, we retain all series and label derived maps explicitly using **coarse semantic buckets** (e.g., `dwi_param_map`, `perf_cbf_like_map`) to avoid forcing the model to separate distinctions that headers fundamentally do not encode. The framework uses (1) heuristic scoring functions to generate high-confidence pseudo-labels, (2) manual verification to create a training dataset, (3) a discriminative classifier for node potentials, and (4) a probabilistic factor graph with Loopy Belief Propagation for session-level refinement. Exported BIDS filenames use standard suffixes with `desc-` labels (e.g., `desc-ADC_dwi.nii.gz`) and derivative JSON sidecars to encode map type and provenance. This architecture minimizes annotation burden while maintaining interpretability, domain-expert trust, and BIDS compliance. Crucially, we adopt a **metadata-only, no-voxel-data, CPU-friendly** design aligned with existing BIDS tooling (HeuDiConv, ezBIDS, BIDScoin).

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1 Problem Setting

1.1 Objective

Given a set of DICOM series extracted from clinical archives (e.g., XNAT pipelines), we aim to:

- Infer BIDS-compatible semantic labels at the series level.
- Predict a joint 3-tuple label (d, s, r) where:
  - $d \in \{\texttt{anat}, \texttt{func}, \texttt{dwi}, \texttt{perf}, \texttt{fmap}, \texttt{unknown}\}$ is the BIDS datatype.
  - s is the BIDS suffix (for raw) or coarse bucket (for derived), conditional on d. Examples:
    * Raw: `T1w`, `T2w`, `FLAIR` for `anat`; `bold`, `sbref` for `func`; `dwi` for `dwi`; `asl`, `dsc`, `dce` for `perf`; `phasediff`, `epi` for `fmap`.
    * Derived: `anat_param_map`, `dwi_param_map`, `perf_cbf_like_map`, `func_stat_map`, etc.
  - $r \in \{\text{True}, \text{False}, \text{None}\}$ indicates derived status: True (derived map: ADC, FA, CBF, reformatted), False (raw acquisition), None (uncertain).
- Enforce physics- and protocol-based constraints across series within the same session.
- Retain all series (no exclusion datatype) and properly label derived maps for downstream handling.

This is formulated as structured prediction using a probabilistic graphical model with weakly supervised node potentials.

1.2 Key Challenge

Manual annotation of clinical DICOM series is expensive and does not scale to millions of series across institutions. We leverage:

1. **Weak supervision** from existing heuristic scoring functions to bootstrap a discriminative model.
2. **Manual verification** of heuristic predictions on a subset to create high-quality training data.
3. **Session-level graph constraints** to refine predictions and enforce global coherence.

1.3 Why Coarse Buckets for Derived Series?

Clinical DICOM repositories contain:

- **Raw acquisitions**: primary data suitable for BIDS `rawdata/`.
- **Derived maps**: quantitative maps (ADC, FA, TRACE, CBF, CBV), reformatted images (MPR, MIP, projections), and vendor post-processing outputs.
- **Scouts/localizers**: planning images with thick slices and few volumes.

Previous approaches excluded all derived series. However, many derived maps are scientifically valuable (e.g., ADC for stroke assessment, CBF for perfusion analysis) and should be:

- Kept in the dataset.
- Labeled explicitly as derived ($r = \text{True}$).
- Placed in BIDS derivatives or flagged in JSON sidecars.

The literature shows that many clinical derived maps copy nearly all acquisition headers from their raw sources, differing mainly in `ImageType` tags and sometimes not even that. Parametric map attributes (units, model, scaling) are often missing or non-standard. Header-only models cannot reliably distinguish ADC vs FA vs other DWI maps, or CBF vs CBV vs MTT, when the metadata is incomplete.

By predicting r jointly with a **coarse derived bucket** (instead of forcing exact suffix discrimination), we:

- Avoid training the model on noise and site-specific quirks.
- Provide **honest, calibrated unaries** to the graph (e.g., "high confidence this is some DWI param map").
- Enable nuanced handling: graph + simple header rules refine to specific suffixes where possible; otherwise export conservative BIDS labels with `desc-` fields.

1.4 Design Constraints

1. **Metadata-only, no voxel data**: consistent with existing BIDS tools (HeuDiConv, ezBIDS, BIDScoin) and avoids GPU requirements.
2. **BIDS-compliant output**: all exported filenames use valid BIDS datatypes and suffixes; derived map types encoded via `desc-` labels and derivative JSON.
3. **Scalable and interpretable**: physics-grounded session-level constraints; domain-expert trust; transparent explanations.

2 Series-Level Feature Representation

Each DICOM series consists of slice-level instances. From header metadata, we extract parameter-wise statistics and construct a structured feature vector.

2.1 Feature Families

We extract features across multiple domains (see `SeriesFeatures` schema):

- **Temporal Features**: Number of timepoints/volumes N_{vol}, repetition time TR (median, IQR), echo time TE (median, range), inversion time TI (presence, median), Is3D flag (`MRAcquisitionType` = 3D).

- **Spatial Features**: Voxel dimensions (v_x, v_y, v_z) (median, variance), slice thickness (median, IQR), number of slices N_{slices}, geometry matrix (affine consistency score).

- **Diffusion Features**: Presence of diffusion tags (b-value, gradient direction), number of directions N_{dir}, multi-shell flag.

- **Perfusion Features**: ASL tagging flags (PASL/PCASL), labeling duration, post-label delay, number of control/label pairs, DSC/DCE indicators (dynamic series with contrast timing).

- **ImageType Features**: Binary flags (`ORIGINAL`, `PRIMARY`, `DERIVED`, `MPR`, `MIP`, `PROJECTION`, `REFORMATTED`, `LOCALIZER`), diffusion map flags (`ADC`, `FA`, `TRACE`, `TENSOR`), perfusion map flags (`CBF`, `CBV`, `MTT`, `TTP`), phase/magnitude flags.

- **Sequence Encoding Features**: Flip angle (median), echo train length, echo spacing, phase encoding direction (i, j, k), multiband factor, parallel imaging acceleration, EPI vs non-EPI flag.

- **Contrast Features**: Contrast agent administered flag, injection time relative to acquisition.

- **Text Features**: `SeriesDescription`, `ProtocolName`, `SequenceName` (tokenized and normalized). Curated keyword sets (`ANAT_KEYWORDS`, `FUNC_KEYWORDS`, `DWI_KEYWORDS`, `PERFUSION_KEYWORDS`, `FMAP_KEYWORDS`, `DERIVED_MAP_KEYWORDS`, `LOCALIZER_KEYWORDS`). Binary flags (`has_anat_token`, `has_func_token`, etc.). Optional TF-IDF vectors over normalized text (limited vocabulary, 100–500 terms).

2.2 Engineered Feature Vector

The resulting feature vector $\mathbf{x}_i \in \mathbb{R}^p$ (where $p \approx$ 50–100 after flattening) contains:

- Continuous features: TR, TE, voxel size, slice thickness, etc.
- Categorical (one-hot) encodings: manufacturer, TR bucket, TE bucket.
- Binary indicators: `ImageType` flags, keyword presence.
- Text representations: sparse binary keyword vectors or TF-IDF.

This representation is designed to be:

- **Vendor-agnostic**: relies on standardized DICOM tags.
- **Robust to free-text variation**: keyword normalization and tokenization.
- **Physics-grounded**: includes acquisition parameters known to distinguish sequence types.

3 Heuristic Scoring Functions

We possess hand-crafted scoring functions originally developed for data exploration and quality control. These functions leverage domain knowledge to assign confidence scores for each axis of the joint label.

3.1 Scoring Architecture

3.1.1 Datatype Scoring

For each datatype $d \in \{\texttt{anat}, \texttt{func}, \texttt{dwi}, \texttt{perf}, \texttt{fmap}\}$, we define:

$$\text{score}_d(\mathbf{x}_i, T_i) = s_d \in \mathbb{Z}_{\geq 0}$$

where $\mathbf{x}_i$ is the feature vector and T_i is the set of text tokens. The score accumulates evidence:

- **Positive evidence**: keyword matches, parameter ranges (e.g., $500 \leq \text{TR} \leq 3000$ ms for T1w).
- **Negative evidence**: exclusionary cues (e.g., DWI keywords penalize `anat` score).

3.1.2 Suffix/Bucket Scoring

For each suffix s (or coarse bucket) conditional on datatype d, we define:

$$\text{score}_{s|d}(\mathbf{x}_i, T_i) = s_{s|d} \in \mathbb{Z}_{\geq 0}$$

For raw series: score standard BIDS suffixes (T1w, T2w, FLAIR, bold, dwi, asl, etc.).
For derived series: score coarse buckets:

- anat_param_map, anat_reformat, anat_segmentation, anat_other_derived
- dwi_param_map, dwi_segmentation_or_mask, dwi_other_derived
- perf_cbf_like_map, perf_cbv_like_map, perf_timing_map, perf_segmentation_or_roi, perf_other_derive
- fmap_susceptibility_or_phase_map, fmap_reformat, fmap_segmentation_or_mask, fmap_other_derived
- func_preproc_bold, func_stat_map, func_timeseries_summary, func_segmentation_or_mask, func_other_derived
- unknown_raw, unknown_derived

Coarse Bucket	Semantics	Datatype	BIDS Suffix Space
anat param map	Quantitative anatomical maps (T1/T2/T2* maps, R1/R2* maps, PD, MT-based, qMRI)	anat	qMRI-style suffixes (T1map, T2map, T2starmap, R1map, R2starmap, PDmap, MTsat); fallback to structural suffixes (T1w, T2w) with desc-
anat reformat	Reformatted anatomical images (MPR, MIP, projections, slab reformats)	anat	Same base structural suffix as source (T1w, T2w, FLAIR), distinguished via desc- (e.g., desc-mpr_T1w)
anat segmentation	Tissue segmentations, parcellations, region labels, masks	anat	dseg, probseg, mask (e.g., desc-aparc_dseg, desc-brain_mask)
anat other derived	Other derived anatomical images not clearly maps/reformats/segs	anat	Structural suffixes (T1w, T2w, FLAIR) with desc- (e.g., desc-biascorr_T1w)
dwi param map	Scalar diffusion maps (ADC, FA, MD, RD, TRACE, etc.)	dwi	DWI scalar-map suffixes (ADC, FA, MD, RD, trace); fallback to dwi with desc-paramap
dwi segmentation or mask	Masks/segmentations in diffusion space (lesion masks, WM skeletons, ROIs, tracts)	dwi	mask, dseg, probseg (e.g., desc-skeleton_mask)
dwi other derived	Other DWI-derived outputs (FODs, ODFs, etc.)	dwi	Base suffix dwi with desc- (e.g., desc-fod_dwi)
perf cbf like map	Cerebral blood flow–like maps (ASL CBF, DSC/DCE CBF)	perf	Perfusion map suffixes (cbf); ASLPrep-style naming (e.g., desc-basil_cbf)
perf cbv like map	Cerebral blood volume–like maps	perf	CBV-style suffixes (cbv, cbvmap)
perf timing map	Perfusion timing metrics (MTT, TTP, Tmax, etc.)	perf	Timing-related perfusion suffixes (mtt, ttp, tmax); fallback to perf
perf segmentation or roi	Perfusion ROI masks, labeled segmentations	perf	mask, dseg, probseg (e.g., desc-lesion_mask)
perf other derived	Other perfusion derivatives (QC images, SNR maps)	perf	Base perf or specific perf suffix with desc-
fmap susceptibility or phase map	B0/B1 field maps, phase maps, susceptibility maps	fmap	Standard fmap suffixes (fieldmap, phasediff, phase1); BEP1/qMRI suffixes (B1map, TB1TFL)
fmap reformat	Reformatted fieldmaps (resampled, unwrapped, projections)	fmap	Same fmap suffix as source with desc-
fmap segmentation or mask	Masks/segmentations in fieldmap space	fmap	mask, dseg, probseg
fmap other derived	Other fieldmap-derived outputs	fmap	Base fmap suffix with desc-
func preproc bold	Preprocessed BOLD time series (motion-corrected, normalized)	func	Suffix bold in derivatives
func stat map	Statistical maps from fMRI (t/z/beta/contrast maps)	func	<source_suffix>map family with stat-<type>
func timeseries summary	Summary maps/timecourses (ALFF, REHO, mean/sd maps)	func	boldmap or bold with stat- or desc-
func segmentation or mask	Functional-space masks or segmentations	func	mask, dseg, probseg
func other derived	Other functional derivatives (confounds, QC images)	func	Suffix bold for images with desc-
unknown raw	Raw-like series with unclear datatype/suffix	unknown	Generic raw suffix or route to sourcedata/
unknown derived	Derived series with unclear map type/datatype	unknown	Generic datatype suffix with desc-unknown

Table 1: Coarse bucket definitions for derived series.

3.1.3 Derived Status Scoring

For derived/raw classification:

$$\text{score}_{\text{derived}}(\mathbf{x}_i, T_i) = s_r \in \mathbb{Z}$$

This score can be negative (evidence for raw) or positive (evidence for derived).

3.2 Conversion to Probabilities

Raw integer scores are converted to probabilities via softmax with temperature scaling:

- **For Datatype**: Compute scores for all datatypes, then

$$P(d = k \mid \mathbf{x}_i) = \frac{\exp(s_k/\tau)}{\sum_\ell \exp(s_\ell/\tau)}$$

  where τ is a temperature parameter (default 1).
- **For Suffix/Bucket**: Conditional on predicted datatype d, compute scores within that datatype:

$$P(s = j \mid d, \mathbf{x}_i) = \frac{\exp(s_j/\tau)}{\sum_{j' \in \mathcal{S}(d)} \exp(s_{j'}/\tau)}$$

  where $\mathcal{S}(d)$ is the set of valid suffixes/buckets for datatype d.
- **For Derived Status**: Use logistic transformation:

$$P(r = \text{True} \mid \mathbf{x}_i) = \frac{1}{1 + \exp(-s_r/\tau)}$$

3.3 Confidence and Labeling

For each axis, we compute:

- Probabilities: $P(d)$, $P(s \mid d)$, $P(r)$.
- Predicted label: $\hat{y} = \arg\max_y P(y)$.
- Margin/confidence: $c = P(\hat{y}) - P(y_{\text{second}})$ (difference between top two probabilities).

High margin ($c \geq 0.7$–0.8) indicates confident predictions suitable for pseudo-labeling.

4 Weak Supervision: Training Data Generation

Rather than treating heuristic scores as fixed node potentials, we use them as weak teachers to generate training data for a discriminative model.

4.1 Pseudo-Labeling Strategy

Step 1: Run Heuristics on Large Corpus

Apply scoring functions to $N \approx 100$k unlabeled series, producing:

- Predicted labels: (d_i, s_i, r_i).
- Confidence scores: $c_d^{(i)}, c_s^{(i)}, c_r^{(i)}$ (margins).

Step 2: Manual Verification for Training Data

1. **Stratified sampling**: Sample series across datatype/confidence bins to ensure coverage:
   - High-confidence predictions ($c \geq 0.8$): 200–300 series per datatype.
   - Medium-confidence predictions ($0.5 \leq c < 0.8$): 100–150 series per datatype.
   - Low-confidence predictions ($c < 0.5$): 50–100 series (focus on edge cases).
2. **Manual review**: Domain expert reviews `SeriesDescription`, `ProtocolName`, and key metadata (TR/TE, `ImageType`, num timepoints) to assign ground-truth labels (d, s, r).
3. **Quality control**: Flag ambiguous cases for discussion or exclusion from training.

Expected manual annotation effort: 500–1000 series across all datatypes and confidence levels.
Step 3: Training Dataset Construction
Combine:

- High-confidence pseudo-labels: $\{(\mathbf{x}_i, d_i, s_i, r_i) \mid c_{\text{joint}}^{(i)} \geq c_{\text{high}}\}$, where $c_{\text{high}} \approx 0.8$.
- Manually verified labels: $\{(\mathbf{x}_j, d_j^*, s_j^*, r_j^*)\}$ from expert review.

This yields:

- Small set of high-quality manual labels: $n_{\text{manual}} \approx 500$–1000.
- Large set of high-confidence pseudo-labels: $n_{\text{pseudo}} \approx 20\text{k}$–50k.

The discriminative model is trained on this combined dataset, with optional weighting (higher weight for manual labels).

4.2 Rationale

- **Data efficiency**: Manual effort focused on 500–1000 series instead of 100k.
- **Coverage**: Pseudo-labels provide breadth across vendors, protocols, and edge cases.
- **Quality**: Manual labels ensure correctness on representative and challenging cases.
- **Iterative refinement**: As model improves, heuristics can be updated or deprecated.

This is a standard weak supervision + active sampling approach, widely validated in medical imaging and NLP literature.

5 Discriminative Node Classifier

A supervised classifier learns to predict the joint 3-tuple (d, s, r) from the feature vector $\mathbf{x}_i$.

5.1 Multi-Task Architecture

We formulate this as multi-task classification with three parallel heads:

$$\begin{aligned} &\mathbf{x}_i \rightarrow \text{shared encoder} \rightarrow \mathbf{h}_i \\ &P_{\text{local}}(d \mid \mathbf{x}_i) \quad \text{(datatype head)} \\ &P_{\text{local}}(s \mid \mathbf{x}_i) \quad \text{(suffix/bucket head)} \\ &P_{\text{local}}(r \mid \mathbf{x}_i) \quad \text{(derived head)} \end{aligned}$$

Loss Function:

$$\mathcal{L} = \mathcal{L}_{\text{datatype}} + \lambda_s \mathcal{L}_{\text{suffix}} + \lambda_r \mathcal{L}_{\text{derived}}$$

where $\mathcal{L}_{\text{datatype}}$ and $\mathcal{L}_{\text{suffix}}$ are cross-entropy losses, $\mathcal{L}_{\text{derived}}$ is binary or 3-class cross-entropy, and λ_s, λ_r are task weights (default 1).

5.2 Model Candidates

We have not yet finalized the classifier architecture. Candidates under consideration:

- **Tree-Based Models (Baseline)**: XGBoost or LightGBM with multi-task output.
  - *Pros*: Robust to mixed numeric/categorical features; handles sparse text features; fast training; interpretable feature importance.
  - *Cons*: Limited modeling of feature interactions beyond trees; no inherent text understanding.
- **Neural Multi-Task Networks**: Shared MLP (2–3 hidden layers, 128–256 units) + three output heads.
  - *Pros*: Can learn complex feature interactions; flexible architecture.
  - *Cons*: Requires careful tuning; may overfit on small manual set without regularization.
- **Multimodal Fusion Models**: Two-tower architecture:
  - *Numeric tower*: MLP on normalized acquisition parameters (TR, TE, voxel size, etc.).
  - *Text tower*: Lightweight transformer (DistilBERT or BioClinicalBERT fine-tuned) or CNN over protocol text.
  - *Fusion*: Concatenate embeddings $\rightarrow$ shared layers $\rightarrow$ multi-task heads.

  - *Pros*: Explicitly models text semantics; can capture nuanced protocol naming.
  - *Cons*: More complex; higher computational cost; requires more data for text tower.
- **Tabular Foundation Models**: TabNet, FT-Transformer, or SAINT.
  - *Pros*: State-of-the-art on tabular benchmarks; built-in feature selection.
  - *Cons*: Less interpretable than trees; requires hyperparameter tuning.

5.3 Training Strategy

Nested Cross-Validation to evaluate feature sets and model architectures rigorously:

- Outer CV: 5 folds for generalization estimate (session-stratified).
- Inner CV: 3–5 folds for hyperparameter tuning and feature selection.

Feature Selection (Optional): Use consensus feature selection across inner CV folds to identify stable, generalizable features. Avoid double-dipping: feature selection must be nested within CV.

Evaluation Metrics:

- Per-axis accuracy and F1: $\mathrm{Acc}(d)$, $\mathrm{Acc}(s \mid d)$, $\mathrm{Acc}(r)$.
- Joint label accuracy: $\mathrm{Acc}(d, s, r)$.
- Confidence calibration: reliability diagrams.
- Session-level completeness: fraction of sessions with all key series labeled.

5.4 Output

The trained classifier provides **calibrated probabilities**:

$$P_{\text{local}}(d, s, r \mid \mathbf{x}_i)$$

which serve as node potentials in the probabilistic graph.

Calibration: Use temperature scaling or Platt scaling to calibrate output probabilities on dev set. This is critical for providing honest unaries to the graph.

6 Probabilistic Study Graph

Neuroimaging acquisitions exhibit structured dependencies that motivate modeling series within a session jointly rather than independently.

6.1 Rationale from Neuroimaging Literature

- Fieldmaps correct susceptibility distortions in EPI and are acquired immediately before/after target runs.
- SBRef images precede BOLD runs for motion correction and co-registration.
- Diffusion acquisitions consist of multi-shell or multi-direction sequences sharing geometry.
- Anatomical scans provide spatial reference for functional normalization.
- Perfusion series (ASL, DSC) often follow anatomical localizers and share timing/geometry constraints.

These relationships are physics- and protocol-grounded and should be encoded as graph constraints.

6.2 Graph Construction

We define a graph $G = (V, E)$ per session:

Nodes V:

- One node per DICOM series i in the session.
- Random variable $Y_i \in \mathcal{L}$ representing the joint label (d, s, r).

Edges E: Edges are only constructed within-session to respect study structure.

Edge types:

1. **Same-Session Membership**: All nodes in a session form a connected component.
   - *Rationale*: Ensures inference operates within coherent acquisition contexts; prevents cross-subject contamination.
2. **Temporal Adjacency**: Connect series acquired within a time window (e.g., $\Delta t \leq 10$ minutes).
   - *Rationale*: SBRef typically precedes BOLD; fieldmaps are often acquired immediately before/after target runs.
3. **Geometry Similarity**: Connect series with similar affine matrices or slice prescriptions (e.g., similarity ≥ 0.9).

- *Rationale*: Fieldmaps and BOLD share slice prescription; SBRef and BOLD share identical geometry.

4. **Fieldmap-Target Pairing**: Based on `PhaseEncodingDirection`, `EchoTime`, `TotalReadoutTime` matching.
   - *Rationale*: Fieldmaps must match target EPI geometry and encoding parameters for valid distortion correction.
5. **SBRef-BOLD Coupling**: Explicit edge between candidate SBRef (single volume, EPI geometry) and BOLD runs.
   - *Rationale*: Strong positive compatibility; SBRef serves as reference for BOLD.
6. **Mutual Exclusion**: Penalize impossible combinations within session.
   - Examples: Two T1w series with identical geometry and timestamp (likely duplicates); derived/localizer images incompatible with primary datatypes.

6.3 Markov Random Field Formulation

Let $\mathbf{Y} = (Y_1, \ldots, Y_N)$ denote all labels in a session. We define a pairwise MRF:

$$P(\mathbf{Y} \mid \mathbf{X}) = \frac{1}{Z} \prod_{i \in V} \psi_i(Y_i \mid \mathbf{x}_i) \prod_{(i,j) \in E} \psi_{ij}(Y_i, Y_j)$$

where:

- $\psi_i(Y_i \mid \mathbf{x}_i)$ = node potential (unary factor) from local classifier.
- $\psi_{ij}(Y_i, Y_j)$ = pairwise potential encoding compatibility constraints.
- Z = partition function (normalization constant).

6.3.1 Factor Graph Implementation

For modularity, we implement this as a factor graph:

- **Variable nodes**: One per series label Y_i.
- **Factor nodes**: Explicit factors for unary and pairwise potentials.
- **Edges**: Connect each factor to the variables it involves (bipartite structure).

This representation makes the factorization explicit and facilitates extension to higher-order constraints (e.g., ternary factors for ASL control/label/M0 triplets).

6.4 Node Potentials

Derived from the weakly supervised discriminative classifier:

$$\psi_i(Y_i = y \mid \mathbf{x}_i) = P_{\text{local}}(d, s, r \mid \mathbf{x}_i)$$

These are calibrated probabilities (validated on dev set) that encode local evidence. **For derived series, s is a coarse bucket rather than exact suffix**, providing honest unaries.

6.5 Pairwise Potentials

For each edge type $(i, j) \in E_{\text{type}}$:

$$\psi_{ij}(Y_i = y_i, Y_j = y_j) = \exp\left(\lambda_{\text{type}} \cdot \phi_{\text{type}}(y_i, y_j)\right)$$

where $\phi_{\text{type}}(y_i, y_j) \in \{-1, 1\}$ is a compatibility score:

- $+1$: strong positive compatibility (e.g., SBRef and BOLD).
- 0: neutral.
- -1: mutual exclusion (e.g., two T1w with same geometry).

and $\lambda_{\text{type}} \geq 0$ controls the strength of the constraint (learned or hand-tuned).

Example Compatibility Matrices:

- **Temporal adjacency**: Encourage $(\texttt{fmap}, \texttt{bold})$, $(\texttt{sbref}, \texttt{bold})$, $(\texttt{anat}, \texttt{perf})$.
- **Geometry similarity**: Strongly encourage $(\texttt{sbref}, \texttt{bold})$ when geometry matches.
- **Fieldmap-target**: Require matching `PhaseEncodingDirection` and compatible TE.
- **Mutual exclusion**: Penalize $(\texttt{T1w}, \texttt{T1w})$ with identical timestamps; penalize $(r = \text{True}, \text{primary datatype})$.

7 Inference via Loopy Belief Propagation

Exact computation of $P(\mathbf{Y} \mid \mathbf{X})$ is intractable for loopy graphs. We use Loopy Belief Propagation (LBP).

7.1 Why LBP?

- Efficient for moderate-sized session graphs (typically 5–20 series per session).
- Approximate marginals via iterative message passing.
- No need for differentiable potentials; works with discrete compatibility matrices.
- Empirically effective for pairwise MRFs in computer vision and structured prediction.

Compared to alternatives:

- **Mean-field**: Often underestimates multimodal interactions; LBP provides better approximations for sparse, loopy graphs.
- **MCMC**: Computationally expensive; overkill for small session graphs.

7.2 Message Passing Algorithm

Message Update:

$$m_{i \to j}(Y_j) = \sum_{Y_i} \psi_i(Y_i) \psi_{ij}(Y_i, Y_j) \prod_{k \in \mathcal{N}(i) \setminus \{j\}} m_{k \to i}(Y_i)$$

Belief (Marginal):

$$b_i(Y_i) \propto \psi_i(Y_i) \prod_{k \in \mathcal{N}(i)} m_{k \to i}(Y_i)$$

Final Label (MAP Estimate):

$$\hat{Y}_i = \arg\max_{Y_i} b_i(Y_i)$$

7.3 Convergence and Scheduling

- **Synchronous updates**: All messages updated in parallel each iteration.
- **Convergence criterion**: $\max_i \|b_i^{(t)} - b_i^{(t-1)}\|_1 < \epsilon$ (e.g., $\epsilon = 10^{-4}$).
- **Max iterations**: Typically 20–50 iterations for small session graphs.
- If LBP does not converge, fall back to:
  - **Damping**: $m^{(t+1)} = (1-\alpha)m_{\text{raw}}^{(t+1)} + \alpha m^{(t)}$, where $\alpha \in [0.5, 0.9]$.
  - **Max-product LBP**: For MAP inference instead of marginals.

8 Assignment of Additional BIDS Entities

After semantic label inference (d, s, r), we assign additional BIDS entities:

8.1 Run Index

Not a classification problem—a session-level ordering problem.

For each session:

1. Group by semantic identity (e.g., all `func-bold` series).
2. Sort by acquisition time.
3. Assign run indices sequentially: `run-01`, `run-02`, etc.

8.2 IntendedFor (Fieldmaps)

Use posterior compatibility scores from the graph:

$$b_i(Y_i = \texttt{fmap}) \cdot \psi_{ij}(Y_i = \texttt{fmap}, Y_j = \texttt{bold})$$

Assign fieldmap to target with highest compatibility.

8.3 Echo, Direction, Acquisition Labels

Derived deterministically from DICOM tags: `EchoNumber`, `DiffusionGradientDirection`, `AcquisitionNumber`.

8.4 Derived Series Handling and BIDS Export

Internal label (d, s, r) where s is a coarse bucket for derived series.

BIDS export strategy:

1. If r = False (raw): Place in `rawdata/` with standard BIDS suffix.
2. If r = True (derived):
   - Use datatype folder from d (e.g., `dwi/`, `perf/`, `func/`).
   - Choose BIDS-valid **base suffix**:
     - If qMRI-BIDS or other extension defines a specific suffix (e.g., `T1map`, `CBF`), use that.
     - Otherwise, use the parent datatype suffix (e.g., `dwi`, `asl`, `T1w`).
   - Encode derived map type in `desc-` label:
     - Examples: `desc-ADC_dwi`, `desc-FA_dwi`, `desc-CBF_asl`, `desc-zstat_bold`.
   - Add derivative JSON sidecar with:

- `Description`: free-text explaining what the map is.
    - `Sources`: BIDS URIs for raw files used to create the derivative.
    - qMRI or other BEP-specific fields (units, model, scaling) where applicable.
  - Where graph + header rules confidently refine coarse bucket to specific map type, emit exact `desc` (e.g., `desc-ADC`).
  - Where confidence is low, use generic but valid label (e.g., `desc-paramap`) or route to `sourcedata/`.
3. If $r =$ None (uncertain): Manual review or default to `sourcedata/` or derivatives for safety.

9 Training & Validation Pipeline

9.1 Phase 1: Heuristic Scoring + Manual Verification

1. Run datatype/suffix/derived scoring functions on large unlabeled corpus ($\approx$ 100k series).
2. Compute confidence scores (margins) for each axis.
3. Stratified sampling: Sample 500–1000 series across datatype/confidence bins.
4. Manual review: Domain expert assigns ground-truth labels (d, s, r).
5. Analyze heuristic performance:
   - Compute accuracy vs. confidence (calibration curves).
   - Identify systematic errors (e.g., `perf` vs `anat` confusion).

9.2 Phase 2: Discriminative Classifier Training

1. **Training data**:
   - High-confidence pseudo-labels (margin ≥ 0.8): 20k–50k series.
   - Manually verified labels: 500–1000 series.
2. **Nested cross-validation**:
   - Outer CV: 5 folds, session-stratified $\rightarrow$ Evaluate generalization.
   - Inner CV: 3–5 folds $\rightarrow$ Hyperparameter tuning, feature selection.
3. **Model selection**: Compare tree-based (XGBoost), neural (MLP), and multimodal (two-tower) models. Evaluate on held-out sessions from multiple institutions and protocols.
4. **Calibration**: Use temperature scaling or Platt scaling to calibrate output probabilities on dev set.

9.3 Phase 3: Graph Inference Tuning

1. Freeze node classifier.
2. Tune pairwise potential weights λ_{type} via:
   - Grid search on validation sessions.
   - Objective: Maximize session-level label consistency and accuracy.
3. Evaluate full pipeline on test sessions:
   - Per-axis accuracy: Acc(d), Acc$(s \mid d)$, Acc(r).
   - Joint label accuracy: Acc(d, s, r).
   - Session-level completeness: fraction of sessions with all key series labeled.
   - Fieldmap-target matching accuracy.

9.4 Phase 4 (Optional): Active Learning

If annotation budget allows or generalization requires improvement:

- Implement active learning: Select sessions with high uncertainty or graph-node disagreement for manual annotation.
- Expand to full weak-supervision label model (Snorkel-style) if additional labeling functions are identified.
- Explore graph-based semi-supervised learning for cross-session label propagation.

10 Why This Architecture?

10.1 Data Efficiency

Weak supervision with pseudo-labeling dramatically reduces manual annotation burden. Empirical evidence from medical imaging and weak-supervision literature shows that pseudo-labeling and label models can approach fully supervised performance with far fewer ground-truth labels. Manual effort focused on 500–1000 representative series instead of 100k.

10.2 Interpretability and Trust

Physics- and protocol-grounded session-level constraints provide transparent explanations for label corrections (e.g., "SBRef reassigned due to geometry mismatch with BOLD"). Domain-expert confidence in outputs via explicit encoding of neuroimaging acquisition logic. Robustness to local classifier errors through global session consistency.

10.3 Modularity and Extensibility

Factor graph implementation separates:

- Node potentials (classifier).
- Edge types (graph structure).
- Compatibility functions (domain rules).

This enables iterative improvement without full system redesign:

- Node model can improve independently via better features, architectures, or training data.
- Graph structure and pairwise potentials can evolve as new protocols or constraints are identified.

10.4 Scalability

Inference is per-session; sessions are independent. Scales to millions of series via session partitioning and parallel processing. Loopy BP on small session graphs (5–20 nodes) is computationally efficient.

10.5 Handling Derived Series

Unlike previous approaches that excluded all derived series, we retain scientifically valuable maps (ADC, FA, CBF) with proper labeling. Enables downstream users to:

- Filter derived series if analyzing only raw data.
- Include derived series for clinical or quantitative analyses.
- Track provenance and processing history via BIDS derivatives JSON.

10.6 Metadata-Only, BIDS-Compliant Design

- Consistent with existing BIDS tooling (HeuDiConv, ezBIDS, BIDScoin).
- No GPU requirement; CPU-friendly for large-scale ingestion pipelines.
- All exported filenames use valid BIDS datatypes and suffixes; derived map types encoded via `desc-` labels and derivative JSON.

11 Open Questions & Design Choices

The following are not yet finalized and require empirical validation:

11.1 Classifier Architecture

Tree-based (XGBoost/LightGBM) vs. neural (MLP) vs. multimodal (two-tower)?
Trade-off: Simplicity & interpretability (trees) vs. expressiveness (neural).
Decision criterion: Nested CV performance on held-out sessions.

11.2 Text Representation

Binary keyword flags (simple, robust) vs. TF-IDF (captures term importance) vs. transformer embeddings (semantic understanding)?
Trade-off: Data efficiency & interpretability vs. expressiveness.
Hypothesis: Keyword flags sufficient for most cases; TF-IDF or transformers may help on rare protocols.

11.3 Graph Potential Learning

Hand-tuned weights (interpretable, fast) vs. learned weights (data-driven, optimal)?
If learned: grid search, Bayesian optimization, or gradient-based learning?
Decision criterion: Ablation study comparing hand-tuned vs. learned on validation sessions.

11.4 Inference Algorithm

Loopy BP (marginals) vs. max-product BP (MAP) vs. mean-field (fast approximation)?
Trade-off: Approximation quality vs. computational cost.
Hypothesis: LBP sufficient for small session graphs; mean-field may suffice if speed is critical.

11.5 Handling Uncertainty

How to handle series where all three axes have low confidence?
Options:

- Flag for manual review.
- Assign to `unknown` category.
- Use graph to propagate confidence from neighbors.
- Route to `sourcedata/` with conservative naming.

11.6 Suffix Refinement from Coarse Buckets

For which derived buckets is it worth attempting suffix refinement (graph + header rules)?

- **Good candidates**: `perf_cbf_like_map` / `perf_cbv_like_map` / `perf_timing_map` (if `SeriesDescription` tokens exist); `fmap` role disambiguation via geometry + PE direction.
- **Hard cases**: `dwi_param_map` (ADC vs FA vs others), `func_stat_map` (t vs z vs beta).

Decision: Define thresholds and deterministic rules for high-confidence refinement; otherwise stay at coarse bucket + `desc-paramap`.

12 Core Conceptual Summary

This framework combines weakly supervised discriminative learning with structured probabilistic reasoning:

1. Heuristic scoring functions generate high-confidence pseudo-labels and confidence scores for a large corpus.
2. Manual verification on a stratified sample (500–1000 series) creates high-quality training data.
3. A discriminative node classifier (architecture TBD: tree-based, neural, or multimodal) learns to predict joint 3-tuple labels (d, s, r) from features, where:
   - s is full BIDS suffix for raw series.
   - s is coarse bucket for derived series (e.g., `dwi_param_map`, `perf_cbf_like_map`).

   Trained on pseudo-labels + manual labels.
4. The node classifier outputs calibrated probabilities that serve as unary potentials in a factor graph.
5. A session-level Markov Random Field (implemented as a factor graph) encodes acquisition physics, protocol regularities, and mutual exclusivity constraints via pairwise potentials.
6. Approximate inference via Loopy Belief Propagation refines predictions to produce globally coherent, interpretable BIDS annotations.
7. All series are retained: Derived series (r = True) are labeled with coarse buckets internally; exported BIDS filenames use valid suffixes + `desc-` labels (e.g., `desc-ADC_dwi`) + derivative JSON for provenance.
8. Metadata-only design (no voxel data, CPU-friendly) ensures scalability and alignment with existing BIDS tooling.

The result is a scalable, vendor-agnostic, and interpretable system for automated DICOM-to-BIDS semantic categorization that:

- Minimizes manual labeling effort (500–1000 series vs. 100k).
- Maintains high accuracy via weak supervision + graph constraints.
- Provides domain-expert trust via transparent, physics-grounded reasoning.
- Enables nuanced handling of derived maps with honest uncertainty quantification.
- Stays BIDS-compliant via standard suffixes + `desc-` labels + derivative JSON.

13 Next Steps

1. Complete manual verification of stratified heuristic predictions (500–1000 series).
2. Benchmark classifier architectures: Compare tree-based (XGBoost), neural (MLP), and multimodal (two-tower) models via nested CV.
3. Implement factor graph: Define edge types, compatibility matrices, and LBP inference.
4. Ablation studies:
   - Heuristics only (no ML).
   - ML nodes only (no graph).
   - Full (graph + ML + graph).
5. Cross-institution validation: Test on held-out sessions from different vendors and protocols.
6. Error analysis: Identify systematic failures and refine heuristics, features, or graph constraints.
7. Define suffix refinement rules: For which coarse buckets and under what confidence thresholds do we attempt fine suffix recovery? Document fallback strategy (`desc-paramap`, `sourcedata/`).

This document serves as a design specification and research proposal. Results, discussion, and future work will be added after empirical validation.